STAT 153 & 248 - Time Series
Lecture Nineteen

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1 $AR(p)$ predictions and Stability

As mentioned previously, predictions for an AR(p) model (given data $y_1, \ldots, y_n$) are obtained by solving the recursion:

$$r_{n+i} = \phi_0 + \phi_1 r_{n+i-1} + \phi_2 r_{n+i-2} + \cdots + \phi_p r_{n+i-p} \qquad \text{for } i = 1, 2, \ldots. \tag{1}$$

which is initialized by

$$r_i = y_i \qquad \text{for } i = n, n-1, \ldots, n+1-p, \tag{2}$$

Here is an interesting observation about the AR(p) prediction recursion (1). Let

$$\mu := \frac{\phi_0}{1 - \phi_1 - \cdots - \phi_p}.$$

Then (1) is equivalent to:

$$r_{n+i} - \mu = \phi_1 \left(r_{n+i-1} - \mu\right) + \phi_2 \left(r_{n+i-2} - \mu\right) + \cdots + \phi_p \left(r_{n+i-p} - \mu\right). \tag{3}$$

One can actually write down a general solution for this recursion. Consider the polynomial:

$$\phi(z) := 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p. \tag{4}$$

This polynomial is of degre p and so it has p roots $z_1, \ldots, z_p$. It turns that for any root z_j of $\phi(\cdot)$ and any constant c,

$$r_i = c z_j^{-i} + \mu$$

solves (1). This is easy to check because if we plug-in $r_{n+i} = c z_j^{-n-i} + \mu$ in (3), we have

$$\textbf{left side of } (3) = c z_j^{-n-i}$$

and

$$\begin{aligned} \textbf{right side of } (3) &= c\left(\phi_1 z_j^{-n-i+1} + \phi_2 z_j^{-n-i+2} + \cdots + \phi_p z_j^{-n-i+p}\right) \\ &= c z_j^{-n-i}\left(\phi_1 z_j + \phi_2 z_j^2 + \cdots + \phi_p z_j^p\right) = c z_j^{-n-i} \end{aligned}$$

because z_j is a root of ϕ so that $\phi_1 z_j + \phi_2 z_j^2 + \cdots + \phi_p z_j^p = 1$.

More generally, by the same logic:

$$r_i = c_1 z_1^{-i} + c_2 z_2^{-i} + \cdots + c_p z_p^{-i} + \mu \tag{5}$$

also solves (3) for any coefficients $c_1, \ldots, c_p$. Because $c_1, \ldots, c_p$ can be chosen so that the initial conditons (2) are satisfied (note that there are p initial conditions in (2) and p coefficients $c_1, \ldots, c_p$), the formula (5) gives the general form of the predictions for the AR(p) model.

The key here is to note that the negative powers of the roots of the polynomial (4) determine the behaviour of the predictions. If the roots are all more than 1 in modulus, then the negative powers would eventually converge to zero, which means that $r_i \to \mu$ as $i \to \infty$. On the other hand, if even one root has modulus strictly smaller than 1, then the predictions (5) can blow up (depending on the values of $c_1, \ldots, c_p$).

The polynomial (4) is called the AR polynomial (note that the AR polynomial is of degree p and it depends on $\phi_1, \ldots \phi_p$, but not on ϕ_0). If all of its roots are strictly larger than 1 in modulus, the AR model is said to be stable. Stability here means that its long-range predictions converge to a constant μ independent of the initialization values $y_{n+1-p}, \ldots, y_n$. The stable regime is also known as 'causal-stationary regime', we will not have time to explain the reasons behind this name.

When we fit an AR(p) using AutoReg, the summary also prints out the roots of the AR polynomial, and their moduli, so that we can quickly assess stability of the fitted AR model.

2 Recursion for AR(p) prediction uncertainties

To compute standard errors associated with AR(p) predictions, we need to calculate:

$$V_i(\theta) := \text{var}\left(y_{n+i} \mid y_1, \ldots, y_n, \theta\right) \qquad \text{for } i = 1, 2, \ldots.$$

The prediction standard error corresponding to the predicted value for y_{n+i} can then be taken to be $\sqrt{V_i(\hat{\theta})}$ (note that θ is replaced by the conditional MLE $\hat{\theta}$).

In the last lecture, we saw that it is difficult to set up a recursion for calculating $V_i(\theta), i = 1, 2, \ldots$. We instead set up the recursion in terms of the covariance matrices:

$$\Gamma_k(\theta) := \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k} \end{pmatrix} \mid \theta, y_1, \ldots, y_n\right)$$

The $(i,j)^{th}$ entry of $\Gamma_k(\theta)$ is

$$\text{Cov}\left(y_{n+i}, y_{n+j} \mid y_1, \ldots, y_n, \theta\right).$$

The diagonal entries of $\Gamma_k(\theta)$ equal $V_1(\theta), \ldots, V_k(\theta)$.

To initialize the recursion for $\Gamma_k(\theta)$, note that

$$\begin{aligned} \Gamma_1(\theta) &= \text{var}\left(y_{n+1} \mid y_1, \ldots, y_n, \theta\right) \\ &= \text{var}\left(\phi_0 + \phi_1 y_n + \ldots \phi_p y_{n+1-p} + \epsilon_{n+1} \mid y_1, \ldots, y_n, \theta\right) \\ &= \text{var}\left(\epsilon_{n+1} \mid y_1, \ldots, y_n, \theta\right) = \sigma^2 \end{aligned}$$

We now relate $\Gamma_{k+1}(\theta)$ to $\Gamma_k(\theta)$ to establish the recursion. We can write

$$\Gamma_k(\theta) = \begin{pmatrix} \Gamma_{k-1}(\theta) & \gamma_{k1}(\theta) \\ \gamma_{k1}^T(\theta) & V_k(\theta) \end{pmatrix}$$

where

$$\gamma_{k1}(\theta) := \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, y_{n+k} \mid \theta, y_1, \ldots, y_n \right)$$

and, as before,

$$V_k(\theta) = \text{var}\left(y_{n+k} \mid \theta, y_1, \ldots, y_n\right).$$

We compute $\hat{\gamma}_{k1}$ as

$$\begin{aligned}
\gamma_{k1}(\theta) &:= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, y_{n+k} \mid \theta, y_1, \ldots, y_n \right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} + \epsilon_{n+k} \mid \theta, \text{data} \right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} \mid \theta, \text{data} \right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \sum_{i=1}^{k-1} a_i y_{n+i} \mid \theta, \text{data} \right)
\end{aligned}$$

where, for $i = 1, \ldots, k-1$,

$$a_i = \begin{cases} \phi_{k-i} & \text{provided } k-p \le i \le k-1 \\ 0 & \text{otherwise} \end{cases}$$

Thus if a is the $(k-1) \times 1$ vector with entries $a_1, \ldots, a_{k-1}$, we have

$$\gamma_{k1}(\theta) = \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, a^T \begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix} \mid \theta, \text{data} \right) = \Gamma_{k-1}(\theta) a.$$

Further

$$\begin{aligned} V_k(\theta) &= \text{var}\left(y_{n+k} \mid \theta, \text{data}\right) \\ &= \text{var}\left(\phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} + \epsilon_{n+k} \mid \theta, \text{data}\right) \\ &= \text{var}\left(\phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} \mid \theta, \text{data}\right) + \sigma^2 \\ &= \text{var}\left(\sum_{i=1}^{k-1} a_i y_{n+i} \mid \theta, \text{data}\right) + \sigma^2 = a^T \Gamma_{k-1}(\theta) a + \sigma^2. \end{aligned}$$

The equation for obtaining $\Gamma_k(\theta)$ from $\Gamma_{k-1}(\theta)$ is therefore

$$\Gamma_k(\theta) = \begin{pmatrix} \Gamma_{k-1}(\theta) & \gamma_{k1}(\theta) \\ \gamma_{k1}^T(\theta) & V_k(\theta) \end{pmatrix} = \begin{pmatrix} \Gamma_{k-1}(\theta) & \Gamma_{k-1}(\theta) a \\ a^T \Gamma_{k-1}(\theta) & a^T \Gamma_{k-1}(\theta) a + \sigma^2 \end{pmatrix}$$

The algorithm for calculating the variances $V_i(\theta)$ for $i = 1, 2, \ldots, K$ is thus given by

1. Initialize with $\Gamma_1(\theta) = V_1(\theta) = \sigma^2$.
2. For $k = 2, 3, \ldots, K$, repeat the following
   a) Form the $(k-1) \times 1$ vector a whose i^{th} entry is ϕ_{k-i} if $k - p \leq i \leq k-1$ and 0 otherwise.
   b) Calculate $\Gamma_k(\theta)$ using $\Gamma_{k-1}(\theta)$ and a by the formula given above.
3. The variances $V_i(\theta), i = 1, 2, \ldots, K$ are given by the diagonal entries of the matrix $\Gamma_K(\theta)$.

Because θ is unknown, in practice, we run this recursion with θ replaced by its conditional MLE $\hat{\theta}$. The prediction standard errors are then the square roots of $V_i(\hat{\theta})$. These prediction errors coincide with those given by the `get_prediction` function in `statsmodels`.

3 MA models

Given a positive integer $q \geq 1$, the Moving Average model with order q (denoted by MA(q)) is defined by the equation:

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} \tag{6}$$

where $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$. The MA($q$) model has $q+2$ unknown parameters which are estimated from observed data: $\mu, \theta_1, \ldots, \theta_q, \sigma$.

The MA(q) model has been called the "Summation of Random Causes" by its inventor Slutzky in the original paper titled "The summation of random causes as the source of cyclic processes" published in Econometrica in 1937. Basically the ϵ_t's can be treated as random causes which are assumed to be independently and identically distributed. The actual observations y_t's are consequences of these causes. The consequence for time t depends on the cause for time t as well as the causes for times $t-1, \ldots, t-q$. These different causes affect the consequence at time t differently depending on the values of $\theta_1, \ldots, \theta_q$. Note that successive observations y_t share some common causes leading to dependence between the successive values of y_t.

Here are some basic properties of the MA(q) model:

1. The mean of y_t is clearly equal to μ (so it is constant in t).

2. The covariance between y_t and y_{t+h} is given by:

$$\begin{aligned}&\operatorname{cov}(y_t, y_{t+h})\\ &= \operatorname{cov}\left(\mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2} + \cdots + \theta_q\epsilon_{t-q}, \mu + \epsilon_{t+h} + \theta_1\epsilon_{t+h-1} + \theta_2\epsilon_{t+h-2} + \cdots + \theta_q\epsilon_{t+h-q}\right)\end{aligned}$$

If $h > q$, it is clear that $\operatorname{cov}(y_t, y_{t+h})$ equals zero. When $0 \leq h \leq q$, we have

$$\operatorname{cov}(y_t, y_{t+h}) = \sigma^2 \sum_{j=0}^{q-h} \theta_j \theta_{j+h}.$$

Thus

$$\operatorname{cov}(y_t, y_{t+h}) = \begin{cases} \sigma^2 \sum_{j=0}^{q-h} \theta_j\theta_{j+h} & : 0 \leq h \leq q \\ 0 & : h > q \end{cases}$$

The simplest of these MA(q) models is MA(1) (i.e., $q = 1$):

$$y_t = \mu + \epsilon_t + \theta\epsilon_{t-1}. \tag{7}$$

For the MA(1) model, we have

$$\text{correlation between } y_t \text{ and } y_{t+h} = \begin{cases} 1 & : h = 0 \\ \frac{\theta_1}{1+\theta_1^2} & : h = 1 \\ 0 & : h > 1 \end{cases}$$

In order to assess whether the MA(q) model is suitable for a particular dataset, a simple way is to compute the sample correlation between y_t and y_{t+h} for each value of h (in other words, the correlation between $(a_t, b_t), t = 1, \ldots, n-h$ where $a_t = y_t$ and $b_t = y_{t+h}$). This was the sample ACF plot that we discussed in the last lecture.

4 On Parameter Estimation in MA(q)

How to fit the parameters of MA(q)? We shall not study this in detail. We will simply rely on the `ARIMA` function for fitting these models to data. But here, I will illustrate the main ideas in the simplest case $q = 1$ i.e., the MA(1) model.

4.1 Full Likelihood for MA(1)

For estimation, the main step is to write down the likelihood. The joint density of $y_1, \ldots, y_n$ is multivariate normal with mean vector $m := (\mu, \ldots, \mu)^T$ and covariance matrix Σ where Σ equals the $n \times n$ matrix whose $(i,j)^{th}$ entry is given by

$$\Sigma(i,j) = \begin{cases} \sigma^2\left(1+\theta^2\right) & \text{when } i = j \\ \sigma^2\theta & \text{when } |i-j| = 1 \\ 0 & \text{for all other } (i,j) \end{cases}$$

The likelihood is therefore

$$\left(\frac{1}{\sqrt{2\pi}}\right)^n (\det\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(y-m)'\Sigma^{-1}(y-m)\right)$$

where y is the $n \times 1$ vector with components $y_1, \ldots, y_n$. This is a function of the unknown parameters μ, θ, σ which can be estimated by maximizing the logarithm of the likelihood. The presence of Σ^{-1} makes this computationally expensive. Some (exact or approximate) formula should be used for Σ^{-1} so that one does not need to invert an $n \times n$ matrix every time the log-likelihood is to be computed.

4.2 Conditional Likelihood for MA(1)

This is an alternative way of obtaining parameter estimates for MA(1).

Instead of the full likelihood, the form of the conditional likelihood when conditioned on $\epsilon_0 = 0$ is simpler. This is similar to the story for the AR(1) model where the form of the likelihood when conditioned on y_1 is much simpler than the full likelihood. The difference here is that instead of conditioning on y_1 being the observed value, we are conditioning on $\epsilon_0 = 0$.

Below we write the conditional likelihood:

$$f_{y_1,\ldots,y_n|\epsilon_0=0}(y_1,\ldots,y_n).$$

This likelihood can be broken down as

$$f_{y_1|\epsilon_0=0}(y_1)f_{y_2|y_1,\epsilon_0=0}(y_2)f_{y_3|y_1,y_2,\epsilon_0=0}(y_3)\cdots f_{y_n|y_1,\ldots,y_{n-1},\epsilon_0=0}(y_n)$$

Each of the terms above can be written explicitly. Let $\hat{\epsilon}_1 = y_1 - \mu$ and, for $t = 2,\ldots,n$,

$$\hat{\epsilon}_t = y_t - \mu - \theta\hat{\epsilon}_{t-1}$$

Then

$$f_{y_1|\epsilon_0=0}(y_1) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y_1-\mu)^2}{2\sigma^2}\right)$$

and

$$\begin{aligned} f_{y_t|y_1,\ldots,y_{t-1},\epsilon=0}(y_t) &= f_{y_t|\epsilon_1=\hat{\epsilon}_1,\epsilon_2=\hat{\epsilon}_2,\ldots,\epsilon_{t-1}=\hat{\epsilon}_{t-1},\epsilon_0=0}(y_t) \\ &= f_{\epsilon_t}(y_t-\mu-\theta\hat{\epsilon}_{t-1}) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1}{2\sigma^2}(y_t-\mu-\theta\hat{\epsilon}_{t-1})^2\right). \end{aligned}$$

Thus the conditional likelihood given $\epsilon_0 = 0$ is given by

$$\begin{aligned} \text{likelihood} &= \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(y_1-\mu)^2}{2\sigma^2}\right)\left[\prod_{t=2}^{n}\frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1}{2\sigma^2}(y_t-\mu-\theta\hat{\epsilon}_{t-1})^2\right)\right] \\ &= \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{\hat{\epsilon}_1^2}{2\sigma^2}\right)\left[\prod_{t=2}^{n}\frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{\hat{\epsilon}_t^2}{2\sigma^2}\right)\right] \\ &= \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n\exp\left(-\frac{1}{2\sigma^2}\sum_{t=1}^{n}\hat{\epsilon}_t^2\right). \end{aligned}$$

The key term in the above likelihood is $\sum_{t=1}^{n}\hat{\epsilon}_t^2$ which depends on the data $\{y_t\}$ and also on the parameters μ and θ. So let us denote

$$S(\mu,\theta) := \sum_{t=1}^{n}\hat{\epsilon}_t^2$$

so that the likelihood and log-likelihood become

$$\text{likelihood} = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^n\exp\left(-\frac{S(\mu,\theta)}{2\sigma^2}\right)$$

and

$$\text{log-likelihood} = -\frac{n}{2}\log(\sigma^2) - \frac{S(\mu,\theta)}{2\sigma^2}.$$

Thus the MLEs of μ, θ are obtained by minimizing $S(\mu, \theta)$:

$$(\hat{\mu}, \hat{\theta}) = \underset{\mu,\theta}{\text{argmin}}\ S(\mu, \theta).$$

We can use numerical optimization techniques to obtain $\hat{\mu}, \hat{\theta}$. For example, in python, scipy has functions for minimization that can be directly used (with some reasonable initialization such at $\mu_{\text{init}} = \bar{y}$ and $\theta_{\text{init}} = 0$).

After obtaining $\hat{\mu}$ and $\hat{\theta}$, the MLE of σ is obtained as

$$\hat{\sigma} = \sqrt{\frac{S(\hat{\mu}, \hat{\theta})}{n}}$$